

RYAN HARTUNG

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EDUCATION

The University of Notre Dame

Ph.D. in Computer Science & Engineering

Notre Dame, IN

Expected May 2029

- Advisor: Dr. Douglas Thain
- Research Focus: Distributed Systems and High-Performance Computing

The University of Notre Dame

Master of Science in Computer Science & Engineering

Notre Dame, IN

August 2025

The University of Texas at Austin

Bachelor of Science in Physics (Computation Option)

Austin, TX

May 2024

- Certificate in Elements of Computing

PUBLICATIONS

[2026-05] Liubov Kurafeeva, **Ryan Hartung**, Benjamin Carter, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Memet Can Vuran, and Rich Wolski. "Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference." *arXiv preprint arXiv:2605.20532*. <https://arxiv.org/abs/2605.20532>

[2025-11] Liubov Kurafeeva, Alan Subedi, **Ryan Hartung**, Michael Fay, Avhishek Biswas, Shantenu Jha, Ozgur Kilic, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture." In *Proceedings of the SC '25 Workshops*. ACM, 2317–2327. DOI: <https://doi.org/10.1145/3731599.3767589>

RESEARCH EXPERIENCE

The University of Notre Dame

Graduate Research Assistant

Notre Dame, IN

January 2025 – Present

- Contributing to the design and implementation of **xGFabric**, architectural framework connecting remote sensor edge networks to HPC centers via 5G.
- Developing low-latency data pipelines to facilitate real-time inference across heterogeneous edge-HPC infrastructure.

The University of Texas at Austin

Undergraduate Research Assistant

Austin, TX

June 2022 – December 2023

- Engineered an open-source Python tool analyzing star-spot patterns across 6,000+ star systems with optimized array processing algorithms.
- Evaluated star-spot contrast ratios using observational data from NASA Kepler and TESS missions to establish baseline metrics for exoplanet detection.
- Integrated parallel processing and multi-core compilation to optimize data throughput and execution runtime.
- Published codebase and workflows to an open-source [GitHub repository](#); research funded by NASA grant.

WORK EXPERIENCE

Hargrove Engineers and Constructors

Controls & Automations Engineering Intern

Katy, TX

May 2024 – August 2024

- Automated the migration of 200,000+ lines of Dow MOD5 code into Honeywell Experion without the loss of functionality.
- Tracked, analyzed, and documented over 7,000 variables to design and draw comprehensive flow diagrams to visualize variable definitions.
- Developed multiple algorithms to expedite the migration and variable mapping processes, improving overall efficiency.

Hargrove Engineers and Constructors

Controls & Automations Engineering Intern

Katy, TX

May 2023 – August 2023

- Instantiated existing library modules to complete the upcoming Reactor and Train configuration for Westlake's PVC plant.
- Performed safety testing on all module and sequence configurations using internal procedures on simulated ABB hardware.
- Successfully configured Modbus communication between the ABB and HIMA system controllers.
- Updated structured text configuration to handle cross-controller communications.

TALKS, PANELS, & PRESENTATIONS

GCASR 2026

Evanston, IL

Poster Presenter

May 11, 2026

- Poster: *Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference*.

Supercomputing '25

St. Louis, MO

Presenter and Demonstrator

November 17 – 19, 2025

- Presented our research and its findings at the Department of Energy's (DOE) booth.
- Performed a live demo in front of an audience of DOE staff, Supercomputing attendees, and fellow researchers at the DOE booth.

Invited Panelist

November 16, 2025

- Participated in an expert panel on edge-to-HPC loop architectures during the XLOOP Workshop [[Recording available online](#)].

HPDC 2025

Notre Dame, IN

Poster Presenter

July 21, 2025

- Poster: *xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture*.

TEACHING EXPERIENCE

The University of Notre Dame

Notre Dame, IN

Graduate Teaching Assistant

August 2024 – May 2025

- Elements of Computing II
Class Size: 37

Spring 2025

Responsibilities:

- Grading
- Office Hours
- Attend class and help students with coding

- Principles of Computing
Class Size: 35

Fall 2024

Responsibilities:

- Grading

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Rust, Java, JavaScript/NodeJS
- **Web Development:** CSS, HTML, JSON, XML
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, Keras
- **Distributed Systems & HPC:** CUDA, HTCondor, OpenMP, SIMD, SLURM, Univa Grid Engine (UGE)
- **Tools, Platforms & Automation:** ABB, Git, Honeywell, L^AT_EX, Linux/UNIX, Microsoft Suite, OpenFOAM, ParaView

LEADERSHIP & SERVICE

Order of Christian Initiation of Adults (OCIA) , Sponsor	September 2026 – Present
Fischer Graduate Residence Mass , Lector	August 2025 – May 2026
Society of Catholic Scientists , Member	August 2025 – Present
Science Policy Initiative at Notre Dame , Member	August 2025 – Present
Catholic Graduate Community , Member	April 2025 – Present
Technology and Catholicism Club , Member	February 2025 – Present
Notre Dame Rock Climbing Club , Member	August 2024 – Present
Notre Dame Texas Club , Member	August 2024 – Present
Science Olympiad , Competitor, Mentor, Club President, Proctor Event-Supervisor, and Lab Coordinator)	August 2011 – Present